Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Releaf

Selected Domain

Domain 2 — Orbyte Smart SME Challenge

Selected SDG Track (1–17)

SDG 8 — Decent Work and Economic Growth

Project / Solution Name

Releaf

Name (Leader)

Lin Jiarun

Email & Phone (Leader)

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University Affiliation(s)

Xiamen University Malaysia

Number of Team Members

5

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small retailers, caterers, bakeries and distributors often coordinate stock, orders and follow-ups across spreadsheets and messages. A draft request can look like a completed movement; manual updates can obscure who changed a quantity and whether goods actually arrived. Missed orders, duplicated entry and unclear task ownership create extra administrative work and make scarce working capital harder to manage.

Our illustrative planning case is a three-person distributor with 120 product lines and 40 stock movements per week. Assuming six minutes of handling per movement gives 240 minutes, or four hours, of weekly administration. These are synthetic assumptions, not an observed customer baseline or measured research result.

Releaf addresses the operational gap between a message or spreadsheet and a reliable business record. Staff need clear document states, permission-controlled changes, traceable approvals and stock balances based on posted movements. Owners also need a simple way to review incoming orders and turn meeting commitments into assigned tasks. The intended economic benefit is less repetitive administration and more reliable small-business operations. Reduced over-ordering and waste are hypotheses to test in a future pilot, not impacts already demonstrated.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Releaf combines inventory workflows, an independent order space and meeting-to-task import with configurable tables and a shared Agent API.

Inventory: staff create inbound or outbound applications, enter product lines manually or import XLSX/CSV, assign an approver and submit. Approval alone does not change stock. Separate posting validates quantities and available stock and records movements atomically. Reversal preserves the original history. Inventory analysis displays movement trends, low-stock alerts and monthly balances.

AI reduces text entry: customer messages become unreviewed orders with no automatically entered amount; meeting summaries become tasks with source quotes; an Assistant calls the same validated operations as the UI. Optional selected-chat WhatsApp intake uses a local bridge, not the official WhatsApp Cloud API. Payments and invoices are human records; orders do not move inventory.

The Assistant defaults to Review first. Explicit Full access does not bypass roles or warehouse workflow rules. Meeting extraction creates tasks for subsequent review, rather than a universal pre-write approval gate. Administrators customize fields while protected system fields preserve business invariants.

The technical approach applies GenAI API integration, strict output validation, relational data, transaction design and deployment configuration. This is an independent Domain 2 prototype: Orbyte summaries may be pasted or imported, but there is no direct Orbyte API or authentication integration.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Runtime: Node.js 22.13+ in a single process
Purpose: Runs the HTTP server and JSON API, using the bundled node:sqlite module.
Frontend: Vanilla JavaScript ES modules, HTML, and CSS
Purpose: Requires no build step and supports switching between English and Chinese.
Database: SQLite via node:sqlite
Purpose: Stores business tables, records, field histories, sessions, and ledger entries.
Data Import and Documents: ExcelJS 4.4.0, csv-parse 7.0.3, and PDF.js 6.4.299
Purpose: Imports XLSX and CSV files and renders PDF pages.
Arithmetic and Rules: Decimal.js 10.6.0 and jsep 1.4.0
Purpose: Provides precise quantity calculations and restricted calculated-field expressions.
Charts: Chart.js 4.6.1
Purpose: Provides locally hosted inventory analysis charts.
Icons: Lucide 1.60.0
Purpose: Supplies icons for the user interface.
AI: OpenAI-compatible Chat Completions, strict json_schema task outputs, and assistant function calling
Purpose: Extracts messages and orders, converts meeting content into tasks, and enables assistant tool usage.
Authentication: Salted scrypt password hashing, HttpOnly session cookies, and administrator/member roles
Purpose: Provides a simulated account and permissions layer, as permitted by the project brief.
Hosting: Local loopback at 127.0.0.1:5174
Purpose: Hosts the current prototype locally; public deployment is pending.

We utilized the official sponsor-provided Cloud Platform Replit. It's used for team colaboration and communication.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design hypothesis is that a clear document lifecycle is more useful to a small team than unrestricted spreadsheet editing. We narrowed the active inventory module to fixed inbound and outbound workflows, separating approval from posting and retaining an immutable movement history.

We iterated using synthetic products, warehouses, transactions, orders and meeting summaries. The demo checks the path from request and spreadsheet import to posted balances and dashboard analysis. Automated tests exercise permissions, validation, rollback, versions and idempotent Agent writes. Mocked AI tests do not demonstrate live extraction accuracy.

The bilingual interface keeps business data unchanged when switching language. Administrators can customize permitted fields; members use protected business workflows. Future validation will involve an SME pilot and baseline measurements. The near-term engineering priority is to complete API business logic and workflow coverage, including dedicated meeting-task Agent operations, consistent errors and stronger permission/retry/failure tests. Public deployment is in progress; production secret management, retention controls and tenant separation remain future work. The team-supplied Replit hosting URL is recorded in the Technical Declaration; public availability still needs verification.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG target 8.3 - support productive activities, entrepreneurship, innovation and the growth of micro-, small- and medium-sized enterprises. Supporting: target 8.2 - productivity through technological upgrading and innovation. Source: United Nations, Goal 8 targets. https://sdgs.un.org/goals/goal8 These are intended contributions, not proof of achieved SDG impact.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Illustrative baseline for planning only: 3 staff, 120 product lines, 40 movements/week and 6 minutes/movement = 240 minutes (4 hours) per week. An 8% monthly product-line discrepancy rate is an assumed comparison scenario, not measured customer data. Establish the actual baseline with an SME before claiming improvement.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	**(1) Minutes per stock-movement document, from first entry to posted; (2) share of posted movements without the assigned approver (target 0); (3) share of product lines whose recorded balance disagrees with the physical count (monthly).**
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed pilot targets, not achieved results: 60% lower handling time relative to the illustrative baseline (6 to 2.4 minutes/movement; 4 to 1.6 hours/week). Zero postings without required approval. Aim for monthly discrepancy at or below 2% of product lines. Validate against the same workload and a physical stock count. No real-world time saving or discrepancy reduction has yet been measured.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Node.js 22 / node:sqlite	Code (runtime)	MIT
ExcelJS 4.4.0; csv-parse 7.0.3; Decimal.js 10.6.0; jsep 1.4.0; Chart.js 4.6.1	Code	MIT
Lucide 1.60.0	Code	ISC + MIT (Feather-derived)
PDF.js (pdfjs-dist) 6.4.299	Code	Apache-2.0

Additional third parties: DeepSeek hosted model/API (deepseek-flash; provider Terms of Use, no team training); optional separate whatsapp-digest-local bridge (MIT) with whatsapp-web.js and qrcode. See Technical Declaration sections 3-4 and repository THIRD_PARTY_NOTICES.md for details.

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

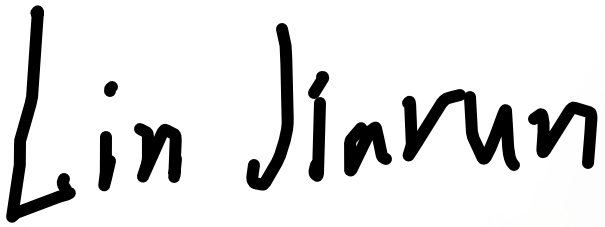
☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

2026-10-25